

i Front matter

Department of Electronic Systems

Examination paper for TFE4141 Design of Digital Systems 1

Examination date: December 4, 2023

Examination time (from-to): 15:00 – 19:00

Permitted examination support material: C / Specified printed and hand-written support material is allowed. A specific basic calculator is allowed.

All printed and hand-written material can be included as part of the allowed specified material.

Academic contact during examination: Per Gunnar Kjeldsberg

Phone: 934 59 550

Academic contact present at the exam location: YES (approximately 16:30)

OTHER INFORMATION

Get an overview of the question set before you start answering the questions.

Read the questions carefully and make your own assumptions. If a question is unclear/vague, make your own assumptions and specify them in your answer. The academic person is only contacted in case of errors or insufficiencies in the question set. Address an invigilator if you suspect errors or insufficiencies. Write down the question in advance.

Hand drawings: For questions **13** and **14** it is possible to submit the entire/some parts of the answer as handwritten sheets. Other questions must be answered directly in Inspira. At the bottom of the question you will find a seven-digit code. Fill in this code in the top left corner of the sheets you wish to submit. We recommend that you do this during the exam. If you require access to the codes after the examination time ends, click "Show submission".

Weighting: The total number of achievable points is 100. The maximum number for each question is given on each question cover page.

Notifications: If there is a need to send a message to the candidates during the exam (e.g. if there is an error in the question set), this will be done by sending a notification in Inspira. A dialogue box will appear. You can re-read the notification by clicking the bell icon in the top right-hand corner of the screen.

Withdrawing from the exam: If you become ill or wish to submit a blank test/withdraw from the exam for another reason, go to the menu in the top right-hand corner and click "Submit blank". This cannot be undone, even if the test is still open.

Access to your answers: After the exam, you can find your answers in the archive in Inspira. Be aware that it may take a working day until any hand-written material is available in the archive.

1 Simulation result (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Given the following VHDL code. Which alternative timing diagram reflects a correct simulation result?

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity task1 is
  port (
    a      : in  std_logic;
    b      : in  std_logic;
    c      : in  std_logic;
    d      : in  std_logic;
    e      : in  std_logic;
    f      : in  std_logic;
    g      : out std_logic);
end task1;

architecture rtl of task1 is
  signal val : std_logic_vector(1 downto 0);
begin
  val <= e & f;
  with (val) select g <=
    a when "00",
    b when "01",
    c when "10",
    d when others;
end rtl;
```

Select one alternative:

Name	Value	0.000 ns	10.000 ns	20.000 ns	30.000 ns
a	0				
b	1				
c	0				
d	1				
e	0				
f	0				
g	0				

Name	Value	0.000 ns	10.000 ns	20.000 ns	30.000 ns
a	0				
b	1				
c	0				
d	1				
e	0				
f	0				
g	0				

Name	Value	0.000 ns	10.000 ns	20.000 ns	30.000 ns
a	0				
b	1				
c	0				
d	1				
e	0				
f	0				
g	0				

Maximum marks: 2

2 Type of circuit (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

What type of circuit does the VHDL code in task 1 implement?

Select one alternative:

- ☐ Half-adder
- ☐ Priority encoder
- ☐ 4x1 multiplexer

Maximum marks: 2

3 Likelihood of faulty chip (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Given the equation $DL = 1 - Y^{(1-T)}$, how likely is it that a customer receives a faulty chip from manufacturing when the percentage of chips produced without any faults is 0.7 and the production testing fault coverage is 0.95?

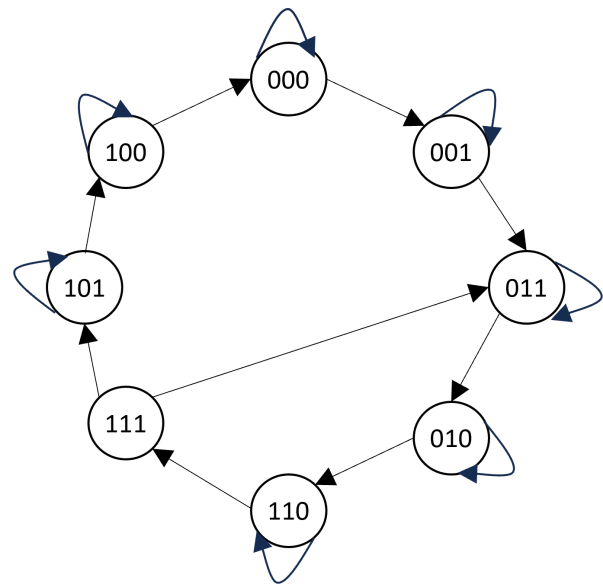
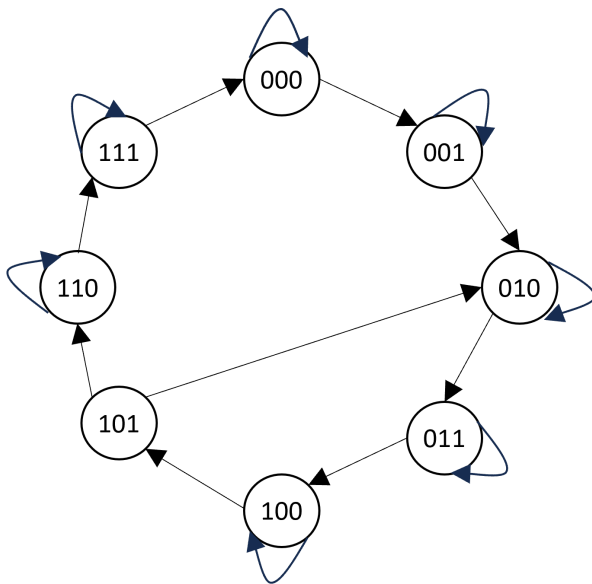
Select one alternative:

- ☐ 1.52%
- ☐ 1.65%
- ☐ 1.77%

Maximum marks: 2

4 State machine coding (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Consider the power consumption of a state machine with the alternative state encodings shown in the state diagram below. We choose to ignore the transition equations now. Which one of the three alternative answers is **correct**?



Select one alternative:

- ☐ The state machine with the encoding to the right has the lowest power consumption.
- ☐ The state machine with the encoding to the left has the lowest power consumption.
- ☐ The state encoding does not influence the power consumption of a state machine.

Maximum marks: 2

5 Combinatorial circuit with registers (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

You have designed a combinatorial circuit with registers on the inputs and outputs. The registers on the input are prepared for clock gating. Which statement is **correct**?

Select one alternative:

- ☐ The clock frequency for the circuit is decided by the delay through the input register, the longest path through the combinatorial circuit, and the delay through the output register
- ☐ If clock gating is enabled, the power consumption in the combinatorial part is given by the equation $P = I_{leak} * V_{dd}$
- ☐ The static power consumption in the circuit will be reduced by using transistors with low threshold voltage in parts of the circuit not along the critical path

Maximum marks: 2

6 Signal declaration (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Given the following signal declaration

signal x std_logic_vector (3 downto 0) := "0101";

Which statement is correct?

Select one alternative:

- ☐ x'low = 3
- ☐ x'low = 0
- ☐ x'low = 1

Maximum marks: 2

7 Model of time (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Given the following VHDL code

```
out1 <= internal1 nor internal2;
```

```
internal1 <= in1 and in2;
```

```
internal2 <= in1 nor in2;
```

and the following event queue:

Time	Signal	Value
T = 0	in1	0
	in2	1
	internal1	U
	internal2	U
	out1	U
T = 0 + Δ	out1	?

What value does **out1** have at time T = 0 + Δ ?

Select one alternative:

☐ 1

☐ 0

☐ U

Maximum marks: 2

8 Entity instantiation (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Given the following entity declaration:

```
entity xor_gate is  
  port(in1, in2: in bit; out1 : out bit);  
end entity xor_gate;
```

and the following three alternative entity instantiations (assuming that there exists an architecture **structure** and signals **in1**, **in2**, and **out1** in the instantiating architecture):

```
DUT1: entity work.xor_gate(structure)  
  port map(in1 => in1, in2 => in2, out1 => out1 );
```

```
DUT2: entity work.xor_gate(structure)  
  port map(out1 => out1, in2 => in2, in1 => in1 );
```

```
DUT3: entity work.xor_gate(structure)  
  port map(in1, in2, out1);
```

Which statement is correct?

Select one alternative:

- ☐ All three instantiations are equivalent
- ☐ The instantiations of DUT1 and DUT3 are equivalent and different from DUT2
- ☐ The instantiations of DUT2 and DUT3 are equivalent and different from DUT1

Maximum marks: 2

9 **Array type (Correct answer 2 points, wrong answer -1 point, no answer 0 points)**

given the following array type declaration and variable instantiation:

```
type matrix is array (0 to 3, 0 to 2) of integer;  
variable picture : matrix :=  
  (0 => (0 => 0, 1 => 2, 2 => 4),  
   1 => (0 => 1, 1 => 3, 2 => 5),  
   2 => (0 => 2, 1 => 4, 2 => 6),  
   3 => (0 => 3, 1 => 5, 2 => 7));
```

What alternative gives the correct value of number after the array accessing **number** <= **picture(1,0)**;

Select one alternative:

- ☐ number = 0
- ☐ number = 2
- ☐ number = 1

Maximum marks: 2

10 **True or false (Correct answer 2 points, wrong answer -1 point, no answer 0 points)**

How many of the following two statements are **correct**?

- A scan chain is used to improve code coverage during design verification.
- Since such a big portion of the digital circuit development time is spent on verification, you as a designer should first and foremost focus on fast code writing when developing a new testbench.

Select one alternative:

- ☐ 0
- ☐ 1
- ☐ 2

Maximum marks: 2

¹¹ Fault model (Maximum 10 points)

Explain what a fault model is and give an example of such a fault model.

Fill in your answer here

Maximum marks: 10

12 Signals and variables (Maximum 10 points)

Describe the difference between signals and variables in VHDL, where they are used and what the benefits are using one instead of the other.

Fill in your answer here

Format	↺	↻	✎	Σ	✖
Words: 0					

Maximum marks: 10

¹³ RSA cryptography (Maximum 16 points)

Assume a «ready valid handshake» protocol as in the term project.

- Explain how the protocol works in the text box below.
- Draw a block diagram that includes a **sender**, that sends data to your **component**, your component itself, and a **receiver** that receives data from your component. The diagram should include all signals required to transfer 32 bit data between sender and your component, and between your component and receiver. Hand in your answer
- Your component needs two clock cycles to complete its calculation on new incoming data from sender before it can be forwarded to receiver. When receiver has received two 32 bit data words, it needs three clock cycles before it can receive more data. Draw a timing diagram with all signals from your block diagram that starts with the receiver and your component being ready to receive data. After two clock periods sender has data to send and is from then on able to send data every clock cycle as long as the timing diagram lasts. Your timing diagram should be at least 20 clock cycles long.

The answers for the two last bullet points should be drawn on separate sheets of paper. These should be marked with the control code and handed in at the end of the exam.

Fill in your answer here

Maximum marks: 16

14 Design task (maximum 44 points)

You are designing a chip that receives 16 bit data from two different sources on a single input bus. We do not know when each of the sources have data to send. At the same time we want to be fair, so that they get an equal opportunity to send if they both have data at the same time. Hence, you need an arbiter. The sources activate their request signal (**R1** and **R0**, respectively) when they want to send, and get a grant signal (**G1** and **G0**, respectively) from your arbiter when they are allowed to send.

When data is received you are going to perform the following calculations

$$Y = Ax^3 + Bx^2 + C, Z = Ax^2 + Bx + D$$

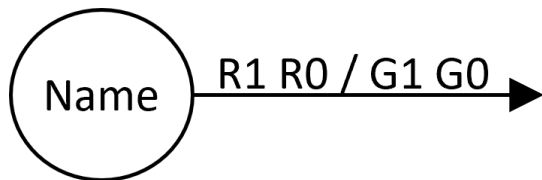
where x is the 16 bit input data and A , B , C , and D are 16 bit constants.

For the following tasks, use the text box below to write as much of your answers as possible (including VHDL code). When necessary, use separate sheets of paper for drawings and other parts not easily included in the text box. These should be marked with the control code and handed in at the end of the exam. Any assumptions you do in your work with the different tasks should be explained in the text box.

a) (max 10 points)

You are going to implement the arbiter as a state machine. Draw a state diagram catching the functionality of the arbiter, including fair granting of data transfer when both sources request access.

Use the following notation in your drawing:



b) (max 8 points)

Write synthesizable VHDL code that implements the state machine.

If

c) (max 10 points)

You are now going to implement a circuit that calculates new values of Y and Z for each new value of x (independent of the source). Show how the calculation is done by drawing a dataflow graph. If possible, rewrite the equations so that you need as few additions and multiplications as possible. How many adders and multipliers do you need?

d) (max 6 points)

You want to estimate how long it takes for the circuit represented in the dataflow graph to calculate Y and Z on a given x value. A multiplier uses 75ns and an adder 35ns for one operation. Assume three cases,

- (i) a fully combinatorial circuit,
- (ii) a circuit with the possibility of having pipeline registers in between the adders and multipliers, and
- (iii) a version that also has the possibility of splitting the multiplier in exactly two with an additional register.

Ignore all delays other than the delay through the adders and multipliers. For each of the cases, indicate both the maximum frequency you get and the total time required to calculate Y and Z .

e) (max 10 points)

Write synthesizable VHDL code that implements the pipelined version of the circuit described by your dataflow graph (case (ii) in task d)).

Fill in your answer here

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Words: 0

Maximum marks: 44